Practical 9

Consider a disk queue with requests forl/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:-

```
  GNU nano 7.2
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12, frames = 3;
    int frame[3];
    int i, j, k, pos, faults = 0, hits = 0;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        if(!found) {
            faults++;
            int farthest = -1;
            pos = -1;

            for(j=0;j<frames;j++) {
                int found_future = 0;

                for(k=i+1;k<n;k++) {
                    if(frame[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            pos = j;
                        }
                        found_future = 1;
                        break;
                    }
                }

                if(!found_future) {
                    pos = j;
                    break;
                }
            }

            if(pos == -1)
                pos = 0;

            frame[pos] = pages[i];
        }
    }

    printf("\nPage Hits = %d", hits);
    printf("\nPage Faults = %d", faults);
    printf("\nHit Percentage = %.2f%%", (float)hits/n*100);
    printf("\nFault Percentage = %.2f%%\n", (float)faults/n*100);

    return 0;
}
```

Output:-

```
Page Hits = 5
Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%
```